

2513830048

[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

May-26-0205

MCA-6203 (Operating System)

MCA 2nd (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from Sections A, B, C and D. Section E is compulsory.

SECTION - A

1. What are different types of multithreading models? Explain advantages and disadvantages of each. (12)
2. What is a Process control block? Explain in detail the mechanisms involved in creating and terminating a process. (12)
3. What is memory fragmentation? (12)

SECTION - B

3. What is thread scheduling? How does it differ from process scheduling? Explain different thread scheduling methods. (12)
4. Explain different CPU scheduling algorithms with examples. (12)

SECTION - C

5. Explain in detail:
- (a) Paging. (2×6=12)
 - (b) Contiguous memory allocation. (2×6=12)
6. What is page replacement? Explain different page replacement algorithms with examples. (12)

2

MCA-6203

SECTION - D

7. What are the various file systems supported by Windows operating system? Explain windows NTFS (NT File System) in detail. (12)
8. Explain how networking in the Linux kernel is implemented. (12)

SECTION - E (Compulsory)

9. Attempt ALL questions from this section. Each question carries 2 marks.
- (a) What are the three main purposes of an operating system?
 - (b) What is the purpose of system calls?
 - (c) Explain Synchronous and asynchronous communication.
 - (d) What do you understand by system boot?
 - (e) What is memory fragmentation?
 - (f) What do you understand by real time operating systems? (6×2=12)